Problem 6: Average Berkeley CS Major
8 Points

Problem ID: `schedule`
Rank: 2

Introduction

Big Ben wants to major in 6-7 majors, including:

- Computer Science,
- Applied Mathematics,
- Rhetoric,
- Theater and Performance Studies,
- East Asian Religion, Thought, and Culture,
- Nuclear Engineering,
- and maybe a Animal Health and Behavior concentration.

However, he is spending too much time studying for his LINGUIS 198: The Art and Science of the Human Voice quiz and doing his Math 98: Cal Pokémon Academy field lab to figure out if he can fit in all the classes he needs to graduate.

Problem Statement

You have $\mathbf{M}$ semesters and can take at most $\mathbf{K}$ courses per semester. There are exactly $\mathbf{N}$ courses offered, each represented by an ID from 1 to $\mathbf{N}$. Some courses have a single corresponding prerequisite course of ID $\mathbf{P}_i$, which must be taken in a semester prior to taking the course of ID i (note that you cannot take a course and its prerequisite simultaneously). If a course doesn't have a prerequisite course, $\mathbf{P}_i = 0$. As an overachiever, you want to take the maximum number of courses possible. Output the maximum number of courses you can take in your $\mathbf{M}$ semesters.

© 2026 CALICO Informatics Competition

Input Format

The first line of the input contains a single integer $\mathbf{T}$ denoting the number of test cases. For each test case:

- The first line contains three space-separated integers $\mathbf{N}$ $\mathbf{M}$ $\mathbf{K}$ where:
  - $\mathbf{N}$ denotes the number of classes.
  - $\mathbf{M}$ denotes the number of semesters.
  - $\mathbf{K}$ denotes the maximum number of classes that can be taken each semester.
- The second line contains $\mathbf{N}$ space-separate integers $\mathbf{P}_1$ $\mathbf{P}_2$ $\ldots$ $\mathbf{P}_\mathbf{N}$, where $\mathbf{P}_i$ denotes the prerequisite course for course i. If there is no prerequisite course for course i, $\mathbf{P}_i = 0$.

Output Format

For each test case, output a single integer representing the maximum number of courses you can take over $\mathbf{M}$ semesters.

Constraints

$1 \leq \mathbf{T} \leq 100$
$1 \leq \mathbf{N} \leq 10^4$
$1 \leq \mathbf{M} \leq 5 \cdot 10^4$
$1 \leq \mathbf{K} \leq \mathbf{N}$
$0 \leq \mathbf{P_i} \leq \mathbf{N}$

It is guaranteed that $\mathbf{P}_i \neq \mathbf{i}$ for all $1 \leq i \leq \mathbf{N}$. Additionally, if you had infinite semesters, it would be possible to take every single course. More formally, there does not exist a sequence of courses $\mathbf{C}_1, \cdots, \mathbf{C}_k$ such that for all $1 \leq i < k$, $\mathbf{C}_i$ is a prerequisite to $\mathbf{C}_{i+1}$ and $\mathbf{C}_k$ is a prerequisite to $\mathbf{C}_1$.

It is also guaranteed that any chain of prerequisites is not longer than 1000. More formally, if $\mathbf{C}_1, \ldots, \mathbf{C}_k$ is a sequence of classes such that $\mathbf{C}_i$ is a prerequisite of $\mathbf{C}_{i+1}$ for $1 \leq i < k$, then $k \leq 1000$.

© 2026 CALICO Informatics Competition

Sample Test Cases

Sample Input Download

```
3
8 3 8
0 1 2 3 4 5 6 7
8 1 8
0 0 0 0 0 0 0 0
8 4 2
0 7 2 3 2 2 1 2
```

Sample Output Download

```
3
8
5
```

Sample Explanations

For the first test case, you must take the courses sequentially in order $1, \ldots, 8$. In 3 semesters, the maximum numbers of courses you can take are 3, when you take courses 1, 2, and 3 each semester in that order. Thus, the answer is 3.

For the second test case, you can take all 8 courses in your first semester. Thus, the answer is 8.

For the third test case, you must take course 1 alone in your first semester (as all other courses have prerequisites), then course 7 alone, then course 2 alone, and then any of 2 courses from courses 3, 5, 6, and 8. Thus, you can take a maximum of 5 courses.

© 2026 CALICO Informatics Competition